AI Vs ML Vs DL Vs DS

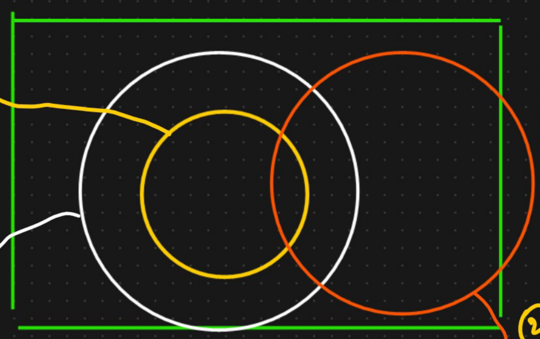
1) Artificial Intelligence
2) Machine Learning
3) Deep Learning
4) Data Science

AI - To create an application which can perform its own task without any human intervention

{Multi Layered NN}

Deep Learning

{ Mimic the Human Brain



ML

Ex : Netflix Recommendation System
→ Action Movies → Action Movies Recommendation

② Self Driving Car

→ Data Science

It provides Stats tools to analyze, visualize, prediction, forecasting the data.

3 Types of Machine Learning.

Machine Learning Techniques

- Supervised ML
- Unsupervised ML
- Reinforcement Learning

Supervised ML Technique

- Regression ✓
- Classification ✓

Predict House price

Dataset

Independent feature

Dependent or o/p feature

Size of House	# of Rooms	Price
5000	5	450K
6000	6	500K
—	—	—
—	—	—

Continuous

① Dependent or o/p feature
② Continuous → Regression
③ Categorical → Classification

Classification

Dependent Feature

No. of study hours	No. of play hours	Pass/Fail
7	3	Pass
2	6	Fail
		May Be

Binary classification

OR

Multiclass classification

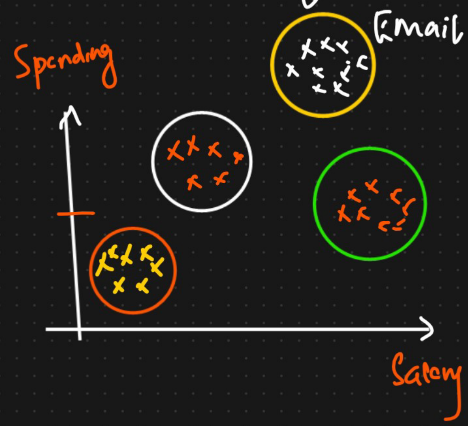
② Unsupervised ML : Eg: Customer Segmentation → Clusters

Salary	Spending Score (1-10)
20000	9
45000	2
—	—
—	—
—	—

Ecommerce Company

⇓

Email with some discount to



Supervised ML

1. Linear Regression
2. Ridge & Lasso
3. ElasticNet
4. Logistic Regression (Classification)
5. Decision Tree
6. Random Forest
7. AdaBoost
8. Xgboost

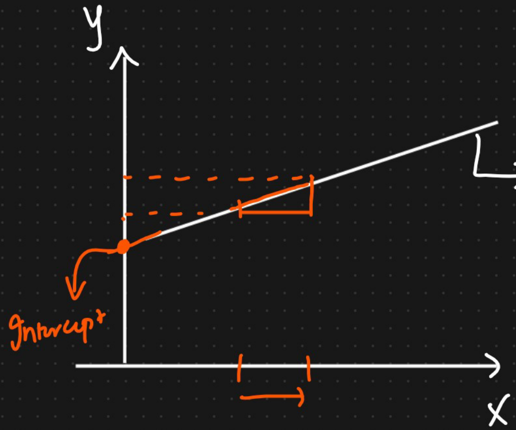
(5–8) Both Classification & Regression.

Unsupervised ML

1. K Means
2. Hierarichel Mean
3. DBScan clustering

Reinforcement Learning

Equation of Line, 3d plane and Hyperplane (n Dimension)



$x = 0$

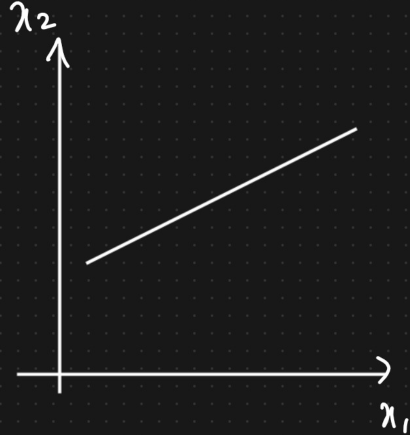
$$y = mx + c \quad \}$$

m = Slope

c = Intercept

$$y = \beta_0 + \beta_1 x$$

$$ax + by + c = 0 \quad \}$$



$$by + c = -ax$$

$$by = -ax - c$$

$$y = \boxed{\frac{-a}{b}}\, x - \boxed{\frac{c}{b}} \rightarrow c$$

$\downarrow$ m

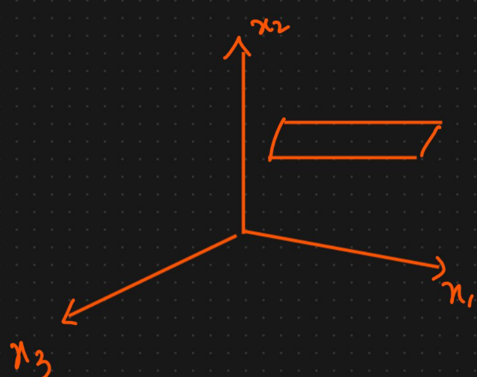
$$y = mx + c \quad \Leftarrow$$

$$w_1 x_1 + w_2 x_2 + b = 0$$

$$\boxed{w^T x + b = 0}$$

$\Downarrow$

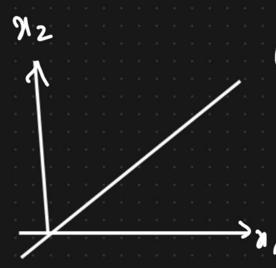
Eq of a Straight line



n-Dimension plane

$$w_1 x_1 + w_2 x_2 + w_3 x_3 + \cdots w_n x_n + b = 0$$

$$\boxed{w^T x + b = 0}$$



$$w_1 x_1 + w_2 x_2 + b = 0$$

$$w_1 x_1 + w_2 x_2 = 0$$

$$\boxed{w^T x = 0}$$

$$w_1 x_1 + w_2 x_2 + w_3 x_3 + b = 0$$

$$\boxed{w^T x + b = 0}$$

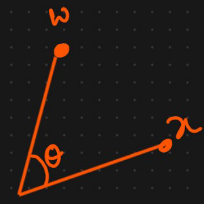
$$w = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} \cdot \; x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Equation of a Straight passing through an origin

$$\boxed{w^T x = 0}$$

Equation of a plane $= \pi_n : \; w^T x = 0$

$$\begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ \vdots \\ w_n \end{bmatrix} \quad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}$$

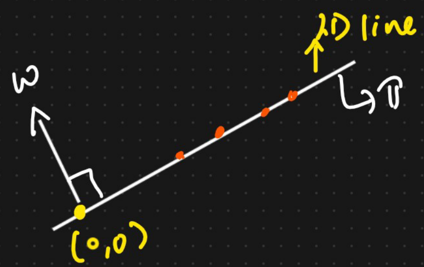


$$w^T x = 0$$

$$w \cdot x = w^T x = \|w\| \, \|x\| \cos\theta = 0$$

$$\theta = 90$$

$$\cos\theta = 0$$



$$\boxed{w \perp \pi}$$

intercept = 0

$$\Downarrow$$

$$w^T x = 0$$

Instance Based Learning Vs Model Based Learning

Usecase

↓

ML Model → Regression / → Classification

Learning pattern

Instance Based | Model Based Technique

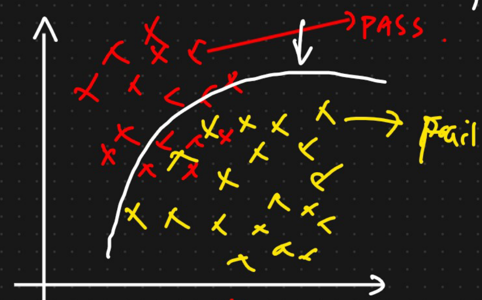
⇓ | ⇓

$\Rightarrow$ DATA → O/p | DATA ⇓ Pattern ⇓ Generalization Method

Learning → Memorizing ⇓ Instance Based | Generalizing ⇓ Model Based Learning

No. of play	No. of Study hours	Pass / Not

{Generalized Model}



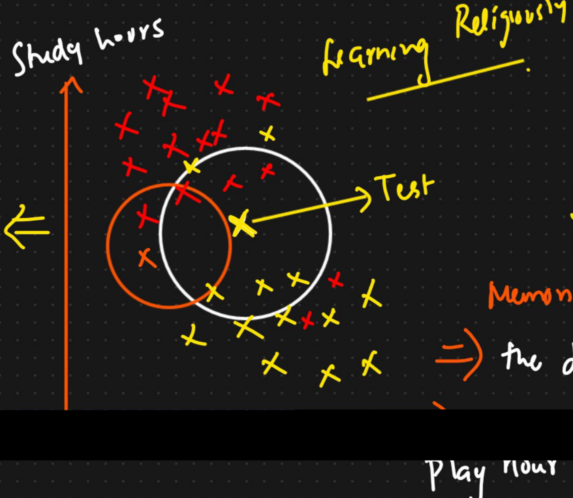
⇓ Generalized Format

Learning Religiously

X → PASS

x → FAIL

① learn pattern of the data



TRAIN ⇐

Instance Based Learning

⇓

Memorizing ⇒ the data. Domain Expert

① KNN {K Nearest Neighbour}

I/p → → Score

X

New

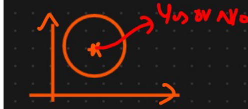
Model

h5 → Kbs or mb

pkl ⇒ hard drive ⇒ Serialized format

Generalization

Decision Boundaries



pkl

I/P → → O/p

Usual/Conventional Machine Learning	Instance Based Learning
Prepare the data for model training ✓	Prepare the data for model training. No difference here ✓
Train model from training data to estimate model parameters i.e. discover patterns	Do not train model. Pattern discovery postponed until scoring query received
Store the model in suitable form	There is no model to store
Generalize the rules in form of model, even before scoring instance is seen	No generalization before scoring. Only generalize for each scoring instance individually as and when seen
Predict for unseen scoring instance using model	Predict for unseen scoring instance using training data directly
Can throw away input/training data after model training	Input/training data must be kept since each query uses part or full set of training observations
Requires a known model form	May not have explicit model form
Storing models generally requires less storage	Storing training data generally requires more storage
Scoring for new instance is generally fast	Storing for new instance may be slow